

ZALAK RAJVANSHI

AI/ML Engineer

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

My experience has been shaped by working through problems from end to end. I've worked across AI and ML, taking projects from the technical problem through to the software and product built around the solution, while picking up new areas as the problem evolves. That has given me real range across AI and product, with the ability to lead, collaborate closely and work independently when the problem calls for it.

Technical Skills

Programming Languages: Python, JavaScript, TypeScript, SQL, HTML / CSS

Machine Learning & AI: Machine Learning, Deep Learning, Computer Vision, NLP, Generative AI, LLMs, RAG, Prompt Engineering, Feature Engineering, Model Training, Model Evaluation, Model Deployment, MLOps

Frameworks & Libraries: TensorFlow, PyTorch, Keras, scikit-learn, Hugging Face Transformers, MediaPipe, OpenCV, LangChain, Pandas, NumPy, React, Next.js, FastAPI, Flask, Node.js, Express.js

Data & Cloud / Tools: SQL, MongoDB, SQLite, Embeddings, Semantic Search, REST APIs, Docker, Git & GitHub, AWS (foundational), CI/CD, Selenium WebDriver, n8n, JWT

Experience

The Product Folks

AI Engineering Intern

Feb 2026 – Jul 2026

Ahmedabad, India

AI Content Engine

Python, Higgsfield MCP, LLM Orchestration, Media Pipelines

- Designed and shipped an AI content production pipeline via **Higgsfield MCP** that converts plain-text briefs into ready-to-publish videos and images. Generated **200+ branded creatives** for community campaigns and marketing at a fraction of the time and cost of a manual creative pipeline.
- Owned the LLM-driven prompt orchestration, automated post-processing and format-aware asset routing across short-form video and image variants, cutting per-asset turnaround from **days to minutes (95% reduction)**.

Semantic Talent Discovery Engine

[Live](#)

Python, Embeddings, Semantic Search, Next.js

- Built an AI candidate-matching platform that ranks best-fit profiles from a **23,000+ indexed professional pool** against any uploaded job description, turning multi-day manual screening into a seconds-long semantic search.
- Designed the embedding pipeline, JD parsing, similarity scoring and filter-aware ranking; deployed as a production Next.js app delivering **top-ranked candidate lists in under 1 second per query** and cutting screening time from **days to minutes per role**.

AI-Driven Lead Generation & Outreach Engine

Python, Selenium, LangChain, LLM

- Developed an end-to-end AI outreach engine combining two modules: the **Lead Generation Module** auto-discovers LinkedIn profiles, parses their career history and ranks them against the active job description; the **Lead Outreach Module** runs a structured **3-message WhatsApp sequence with 36-hour follow-up gaps**, cutting manual sourcing-to-engagement effort by **70%**.
- Added **LLM-powered reply classification** (85% 3-class accuracy across positive / negative / neutral), a daily summary bot for HR visibility and persisted-state recovery; designed to scale to **100+ candidates per campaign** end-to-end without human handoff.

Arova - Conversational AI Platform

[Live](#)

Python, JavaScript, LangChain

- Delivered Arova, a production conversational AI platform powering **24/7 community engagement** across **8-10 high-frequency intent flows**, cutting response latency from **hours to under 2 seconds** and eliminating **80% of manual triage effort** while orchestrating end-to-end conversations without human handoff.
- Composed the system across three layers: a JavaScript conversational UI, a Python **LangChain**-powered backend for intent routing and conversation-memory handling, plus a separate AI processing pipeline for asynchronous inference, with prompt engineering and graceful fallback for low-confidence or out-of-scope queries.

Projects

Sign Language Detection (ASL to Speech)

[GitHub](#)

Python, TensorFlow, MediaPipe, OpenCV, pyttsx3

- Built a real-time **American Sign Language** recognition system that translates **29 gesture classes** into full-sentence speech, enabling accessible communication for non-signers; runs at **25-30 FPS** (sub-40 ms per frame) on a standard CPU with no GPU required.
- Engineered the ML pipeline using **MediaPipe** landmark extraction and a **Conv1D CNN** trained on **30K+ feature frames**, reaching **98.78% validation accuracy**; added confidence-based stability filtering (7-10 frames, over 88% confidence), TextBlob spell correction and pyttsx3 text-to-speech for readable output.

Gitlane - AI Commits, Secret Guard & GitHub Coach

[GitHub](#)

Python, TypeScript, Flask, SQLite, LLM, VS Code Extension API, GitHub API

- Built an AI developer tool spanning a **CLI, browser dashboard and VS Code extension** that generates **Conventional-Commits** messages straight from `git diff`, so contributors ship clean, consistent history without hand-writing messages; released open-source (MIT) with a cross-platform installer.
- Added an automated secret-guard module that scans staged files for leaked API keys and remediates them by moving secrets to `.env` and rewriting source with `os.getenv` / `process.env`, preventing pre-push credential leaks; tracks daily shipping streaks, weekly sprints and per-repo goals through a shared **SQLite** memory layer.

SmartTask AI - Context-Aware Day Planner

[GitHub](#)

React, Vite, Tailwind, Node, Express, PyTorch, Hugging Face Transformers, JWT

- Built a full-stack productivity app that converts natural-language goals into time-blocked, prioritised daily schedules in **under 3 seconds**, replacing the morning planning users would otherwise map out by hand.
- Developed a **React + Vite + Tailwind** frontend wired to a **Node + Express** backend with **PyTorch / Hugging Face Transformers** inference; secured the app with **JWT** authentication and externalised model weights for streamlined AWS / Render deployment.

Smart Notes AI - Handwriting to Structured Notes

[GitHub](#)

Python, OCR, Layout Analysis, JavaScript

- Built a pipeline that converts handwritten pages into searchable, structured digital notes, turning minutes of manual retyping into a few seconds of processing per page.
- Combined **OCR**, layout analysis and semantic classification to reach **85% line-level extraction accuracy**, splitting the system into a Python inference backend and a JavaScript note-management frontend with PDF export.

Education

Silver Oak University (SOCET)

2022 – 2026

B.Tech, Computer Engineering - CGPA: 8.90 / 10

Ahmedabad, India

Certifications

Foundations of AI & ML, Microsoft - Nov 2024 | **Gen AI Study Jams 2024**, Google Cloud Skills Boost - Nov 2024

Leadership & Community

- **Vice Chairperson**, *IEEE SOU WIE SB AG* (2025–26): Led technical and professional initiatives across the chapter, coordinated multi-event execution and championed women-in-engineering programming on campus.
- **Senior Curation Executive**, *IEEE SOU Student Branch* (2026): Curated technical events, speaker sessions and knowledge-sharing activities; owned content quality and run-of-show for chapter programming.
- **Content Lead**, *GDG On Campus SOU* (2024–25): Owned the chapter's technical content strategy and produced developer-facing material that drove community engagement and event attendance.
- **Community Volunteer**, *AWS Cloud Clubs SOU* (2024–Present): Co-organized cloud workshops and hands-on AWS labs, onboarding students into core cloud services.